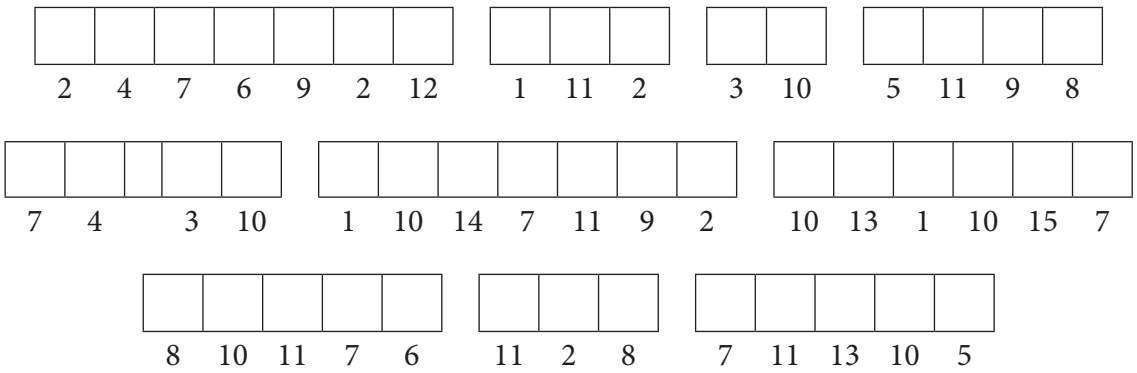
Name: _____

WORKSHEET

Advanced exponential and periodic functions

In this world …



Benjamin Franklin, 1817

Questions

Write an equivalent statement for each of the following.

1 $1 + \tan^2(\theta)$

2 $2\cos^2(2\theta) + 2\sin^2(2\theta)$

3 $\cot^2(\theta)$

4 $\sin(A + B)$

5 $\cot(A - B)$

6 $\cos\left(\frac{\pi}{2} - B\right)$

7 $\operatorname{cosec}\left(\frac{3\pi}{2} + A\right)$

8 $\sin(2A)$

9 $\sec[2(B - \pi)]$

Evaluate the following.

10 $\sin\left(\frac{5\pi}{6}\right)$

11 $\sec\left(\frac{7\pi}{6}\right)$

12 $\cot\left(\frac{13\pi}{6}\right)$

13 $\cos\left(-\frac{13\pi}{6}\right)$

14 $\sin\left(\frac{\pi}{2} + \frac{\pi}{3}\right)$

15 $\cos\left(2 \times \frac{\pi}{3}\right)$

Solutions

A $-\frac{2}{\sqrt{3}}$

B $\operatorname{cosec}^2(\theta) - 1$

C $\sec^2(\theta)$

D $2\sin(A)\cos(A)$

E $\frac{1}{2}$

G $\sqrt{3}$

H $\sin(B)$

I $\sec(2B)$

N 2

O $\sin(A)\cos(B) + \cos(A)\sin(B)$

P $-\frac{1}{2}$

R $\frac{1}{2}$

S $\frac{1 + \tan(A)\tan(B)}{\tan(A) - \tan(B)}$

T $-\sec(A)$

X $\frac{\sqrt{3}}{2}$

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